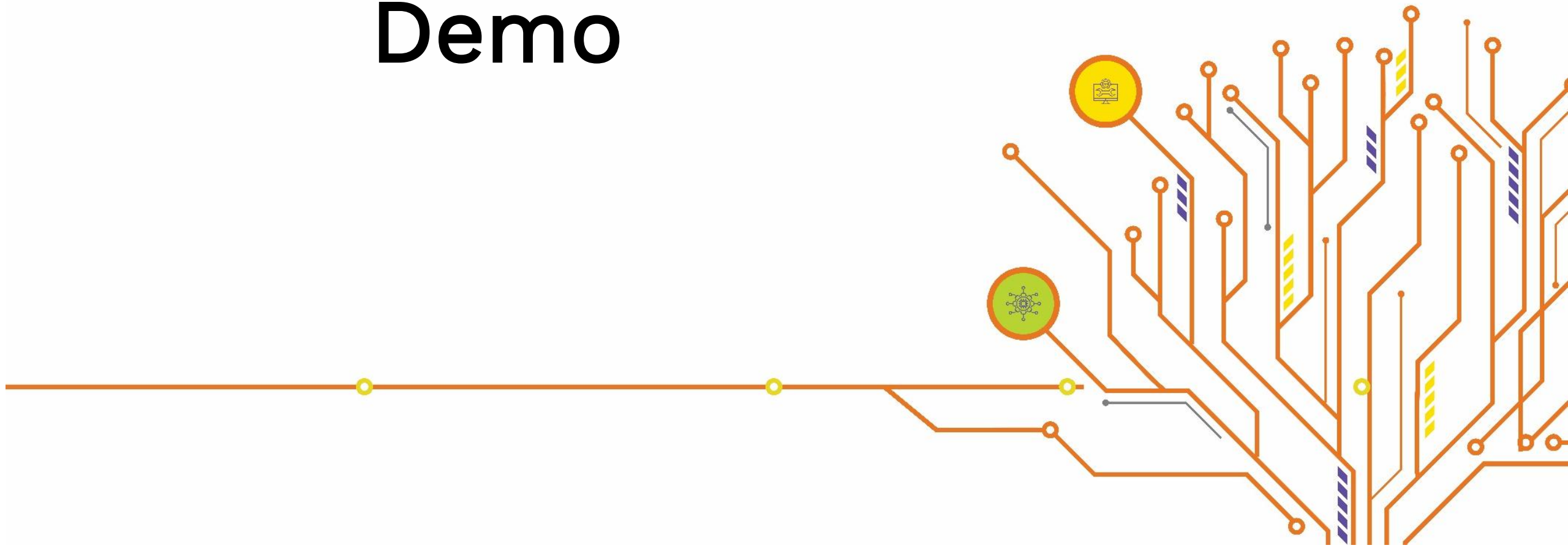
KPIT | APEX LAB

Understanding Demo



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User Programs Access Devices: Syscalls (1/3)

Trace how a user program interacts with a device via syscalls → VFS → driver.

1. Trace syscalls with strace

```
$ cd ~
$ mkdir demo && cd demo
$ strace -o trace.txt -f cat /dev/zero | head -c 1024 > /tmp/junk
$ sed -n '1,60p' trace.txt
```

What this does:

- cat /dev/zero → produces an infinite stream of zero bytes.
- head -c 1024 → consumes only the first 1024 bytes, then exits.
- strace -o trace.txt -f → records syscalls from cat (and its children if any; head isn't a child here, so it won't be traced).
- Output redirected to /tmp/junk.

User Programs Access Devices: Syscalls (2/3)

Terminal will show details like this:

407636 execve("/usr/bin/cat", ["cat", "/dev/zero"], ...) = 0 → User-space program starts.

407636 openat(AT_FDCWD, "/dev/zero", O_RDONLY) = 3 → User space requests open(); kernel VFS routes to /dev/zero driver.

407636 read(3, "\177ELF\2\1\1\3...", 832) = 832 407636 read(3, "\0\0\0\0\0\0\0...", 131072) = 131072 → User space calls read(); kernel returns bytes. For /dev/zero, it *generates zeros*.

407636 write(1, "\0\0\0\0\0\0\0...", 131072) = 65536 → User space calls write(); kernel writes to a pipe (stdout → head).

407636 --- SIGPIPE {si_signo=SIGPIPE, ...} ---

407636 +++ killed by SIGPIPE +++ → head consumed 1024 bytes and closed the pipe; kernel signals cat with SIGPIPE.

Key mapping:

- User space: cat issues open, read, write calls.
- Kernel space: VFS + device driver /dev/zero + pipe implementation handle the actual work.
- write() returning 65536 (not 131072) is due to pipe capacity.

cat --read --> /dev/zero driver
--write --> pipe buffer
head reads 1024 bytes → closes → SIGPIPE to cat

User Programs Access Devices: Syscalls (3/3)

2. Execute

```
grep '/dev/zero' trace.txt        # Show open()
grep 'read(' trace.txt | head     # Show read() calls
grep 'write(' trace.txt | head    # Show write() calls
```

Observation:

```
423993 execve("/usr/bin/cat", ["cat", "/dev/zero"], 0x7ffece84b420 /* 48 vars */) = 0
423993 openat(AT_FDCWD, "/dev/zero", O_RDONLY) = 3
423993 read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0P\237\2\0\0\0\0\0"..., 832) = 832 423993 read(3,
"\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0"..., 131072) = 131072 423993 write(1,
"\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0"..., 131072) = 65536
```

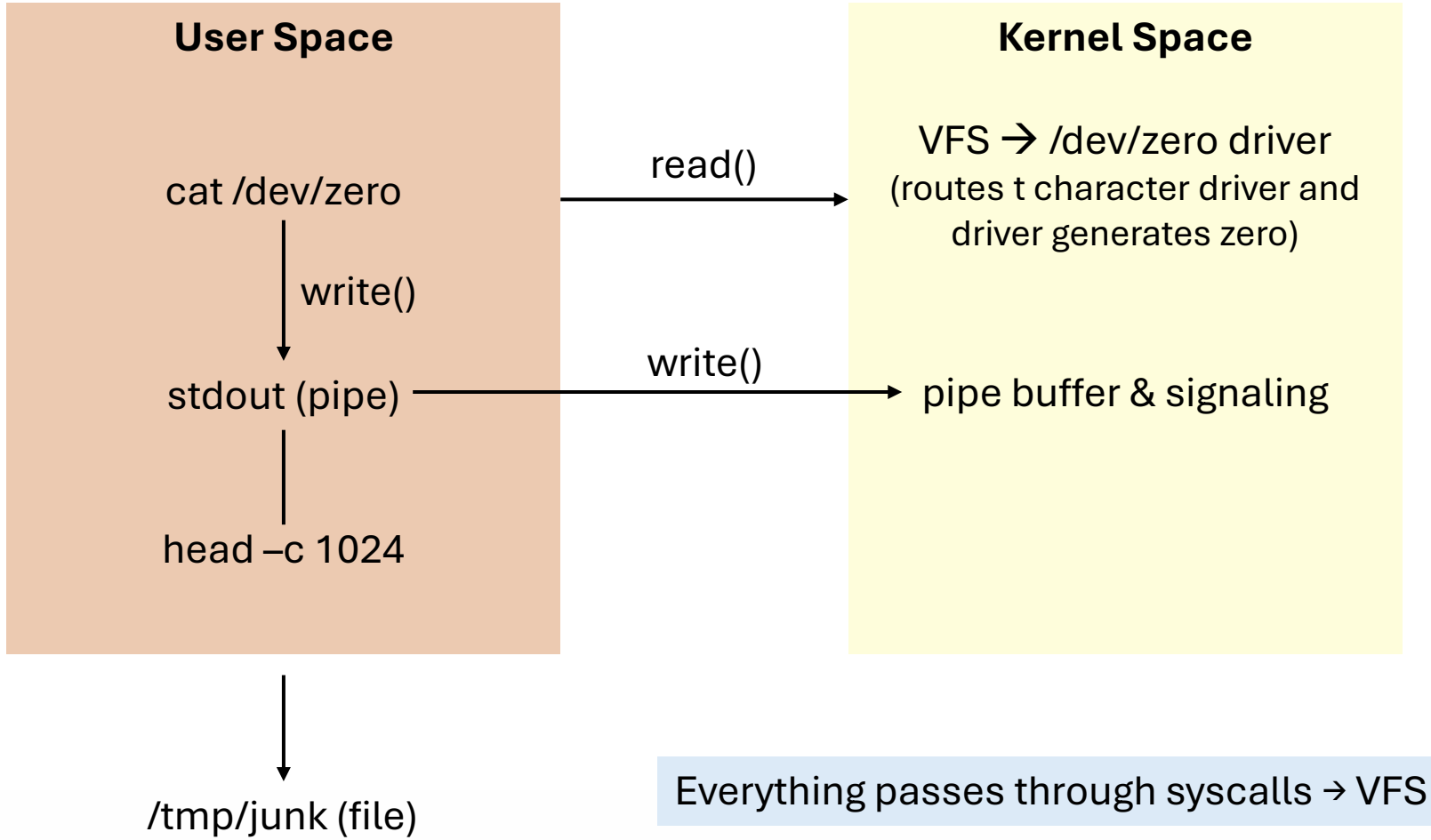
read(3, buf, 131072) = 131072

- 3 → file descriptor (FD) for /dev/zero.
- 131072 (inside) → requested size.
- = 131072 (after) → actual bytes returned by kernel.

write(1, buf, 131072) = 65536

- 1 → FD for stdout (a **pipe** in this pipeline).
- Requested 131072 bytes, kernel actually wrote 65536 (pipe capacity/consumer behavior).

User → Kernel → Device



Everything passes through syscalls → VFS → driver.

Why /dev/zero ≠ Regular File?

Why does /dev/zero behave differently from a regular file?

- /dev/zero is a **character device** exposed via a **special device node**.
- /dev/zero is part of major number 1 → memory devices
- It's backed by a **kernel driver** that returns zero bytes on read **indefinitely**.
- No disk access occurs; it's generated in kernel space.

A regular file has finite size stored on disk; /dev/zero is infinite and virtual.

Kernel Internals via Virtual Filesystems (1/)

/proc & /sys

```
$ uname -a
Linux PL-9L 6.8.0-90-generic #91~22.04.1-Ubuntu SMP PREEMPT_DYNAMIC Thu Nov 20 15:20:45 UTC 2 x86_64 x86_64 x86_64 GNU/Linux
```

- Linux → Kernel name.
- PL-9L → Hostname of your machine.
- 6.8.0-90-generic → Kernel version.
- #91~22.04.1-Ubuntu → Build number and Ubuntu-specific patch level.
- SMP → Symmetric Multi-Processing (supports multiple CPUs).
- PREEMPT_DYNAMIC → Kernel preemption model (dynamic preemption).
- Thu Nov 20 15:20:45 UTC 2 → Kernel build date/time.
- x86_64 x86_64 x86_64 → Architecture: kernel, CPU, and platform are all 64-bit.
- GNU/Linux → Operating system type.

Kernel Internals via Virtual Filesystems

```
$ cat /proc/meminfo
```

- MemTotal: 32605452 kB → Total physical RAM (~32 GB).
- MemFree: 17589048 kB → Free memory not used by anything (~17 GB).
- MemAvailable: 22411248 kB → Memory available for new apps (~22 GB).
- Buffers: 180696 kB → Memory used for block device buffers.
- Cached: 6041868 kB → Memory used for page cache (file data).
- SwapCached: 68 kB → Swap pages cached in RAM.
- Active: 10223932 kB → Memory actively used by processes.
- Inactive: 3445064 kB → Memory not actively used but still allocated.
- Active(anon): 8538800 kB → Anonymous memory actively used (heap, stack).
- Inactive(anon): 75312 kB → Anonymous memory inactive.

Key point: These values are **generated by the kernel on demand**, not stored on disk.

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Kernel Internals via Virtual Filesystems

```
$ cat ls /sys/class
```

accel, block, net, power_supply, thermal, tty, usbmisc …

Explanation:

- Each directory represents a **device class** exposed by sysfs.
- Examples:
    - block → Block devices (disks).
    - net → Network interfaces.
    - thermal → Thermal sensors.
    - tty → Terminals.
    - power_supply → Battery/AC power info.

Kernel Internals via Virtual Filesystems

```
ls /sys/class/net
```

- enp0s31f6 lo virbr0 wlp0s20f3 zcctun0

Explanation:

- enp0s31f6 → Wired Ethernet interface.
- lo → Loopback interface (127.0.0.1).
- virbr0 → Virtual bridge (used by virtualization).
- wlp0s20f3 → Wireless interface.
- zcctun0 → Likely a VPN or tunnel interface.

Is /proc/meminfo stored on disk?

- No, it’s generated by the kernel on demand.

Device Nodes & Major/Minor Numbers

```
ls -l /dev/null /dev/zero /dev/tty
```

```
crw-rw-rw- 1 root root 1, 3 Jan 15 11:20 /dev/null
crw-rw-rw- 1 root tty  5, 0 Jan 15 11:20 /dev/tty
crw-rw-rw- 1 root root 1, 5 Jan 15 11:20 /dev/zero
```

- **Leading c** → Character device (stream-oriented, e.g., /dev/null, /dev/zero, /dev/tty).
- If it were b, it would be a block device (disk-like, e.g., /dev/sda).
- **Permissions (rw-rw-rw-)** → Anyone can read/write these devices.
- **Owner/Group (root root)** → Owned by root.
- **1, 3 or 1, 5** → **Major, Minor numbers**:
- **Major** = Identifies the driver in the kernel (e.g., 1 = memory devices).
- **Minor** = Identifies the specific device handled by that driver (3 = /dev/null, 5 = /dev/zero).
- **Path** → /dev/null, /dev/zero, /dev/tty are **device nodes**, not regular files.

What happens if you delete /dev/null?

- You lose the handle to the driver;
- programs that rely on /dev/null (for discarding output) will fail.
- The driver still exists in the kernel, but the user-space entry point is gone.

Show major/minor mapping

```
cat /proc/devices | head
```

- This lists all registered major numbers and their drivers.

annotated explanation for each entry in your /proc/devices snippet:

1 mem
→ Major number 1 is assigned to yesthe "memory" driver family.
Devices under this include:
- /dev/null (discard data)
- /dev/zero (generate zeros)
- /dev/full, /dev/random, /dev/urandom
These are all character devices that interact with kernel memory.

4 /dev/vc/0
→ Major number 4 is for virtual consoles.
/dev/vc/0 is the first virtual console device (used for text terminals).

4 tty
→ Also under major 4: TTY devices (teletype terminals).
These represent physical or virtual terminal interfaces.

4 ttyS
→ Serial TTY devices (e.g., /dev/ttyS0 for COM ports).
Used for serial communication.

5 /dev/tty
→ Major number 5 is for system TTY devices.
/dev/tty refers to the controlling terminal for the current process.

5 /dev/console
→ The system console device (kernel messages and login prompt).
Critical for system interaction during boot.

5 /dev/ptmx
→ Pseudo-terminal master multiplexer.
Used to create new pseudo-terminal pairs for terminal emulators.

5 ttyprintk
→ Special TTY device that prints kernel messages to a TTY.
Useful for debugging.

6 lp
→ Major number 6 is for line printers.
Historically used for parallel port printers (e.g., /dev/lp0).

Observing a Live Process

```
sleep 1000 &        # Start a process in background
PID=$!              # Capture its PID
ls /proc/$PID
```

- Shows directories and files representing the process state:
- status → Human-readable summary (threads, memory usage, state).
- maps → Virtual memory layout.
- fd/ → Open file descriptors.
- cmdline → Command-line arguments.
- environ → Environment variables.

Observing a Live Process

```
cat /proc/$PID/status | head   # Show process status
```

- Name: sleep
- Umask: 0022
- State: S (sleeping)
- Tgid: 449028
- Ngid: 0
- Pid: 449028
- PPid: 406625
- TracerPid: 0
- Uid: 923518422 923518422 923518422 923518422
- Gid: 1214800513 1214800513 1214800513 1214800513

Observing a Live Process

```
cat /proc/$PID/maps | head
```

Start	End	Perms	Offset	Dev	Inode	Path
5b7999ddc000	5b7999dde000	r--p	00000000	103:02	30147684	/usr/bin/sleep
5b7999dde000	5b7999de2000	r-xp	00002000	103:02	30147684	/usr/bin/sleep
5b7999de2000	5b7999de3000	r--p	00006000	103:02	30147684	/usr/bin/sleep
5b7999de4000	5b7999de5000	r--p	00007000	103:02	30147684	/usr/bin/sleep
5b7999de5000	5b7999de6000	rw-p	00008000	103:02	30147684	/usr/bin/sleep
5b79a2331000	5b79a2352000	rw-p	00000000	00:00	0	[heap]
7903de000000	7903de573000	r--p	00000000	103:02	30148332	/usr/lib/locale/locale-archive
7903de600000	7903de628000	r--p	00000000	103:02	30172945	/usr/lib/x86_64-linux-gnu/libc.so.6
7903de628000	7903de7bd000	r-xp	00028000	103:02	30172945	/usr/lib/x86_64-linux-gnu/libc.so.6
7903de7bd000	7903de815000	r--p	001bd000	103:02	30172945	/usr/lib/x86_64-linux-gnu/libc.so.6

Observing a Live Process

“Why do we see multiple entries for /usr/bin/sleep?”

- Expected answer: Different sections (code, data) with different permissions.

“What does [heap] mean?”

- Expected answer: Dynamic memory region for allocations.

“Why is libc.so.6 mapped?”

- Expected answer: Because sleep uses libc functions.